

Why statistics?

Top 10 Uses Of Statistics In Our Day to Day Life



Predictions



The figures help us make predictions about something that is going to happen in the future. Based on what we face in our daily lives, we make predictions. How accurate this prediction will depend on many factors. When we make a prediction, we take into account the external or internal factors that may affect our future. Doctors, engineers, artists, and practitioners all use statistics to make predictions about future events. For example, doctors use statistics to understand the future of the disease. They can predict the magnitude of the flue in each winter season through the use of data. Engineers use statistics to estimate the success of their ongoing project, and they also use the data to evaluate how long it will take to complete a project.

Quality testing



Companies make thousands of products every day and each company must make sure that a good quality item is sold. But a company can't test each and every item that they ship to you, the consumer. So the company uses statistics to test just a few, called a sample, of what they make. If the sample passes quality tests, then the company assumes that all the items made in the group, called a batch, are good.

Weather forecasting



Have you ever seen weather forecasting? Do you know how the government does the weather forecasting? Statistics play a crucial role in weather forecasting. The computer use in weather forecasting is based on the set of statistics functions. All these statistics function to compare the weather condition with the pre-recorded seasons and conditions. This helps the government.

Emergency Preparedness



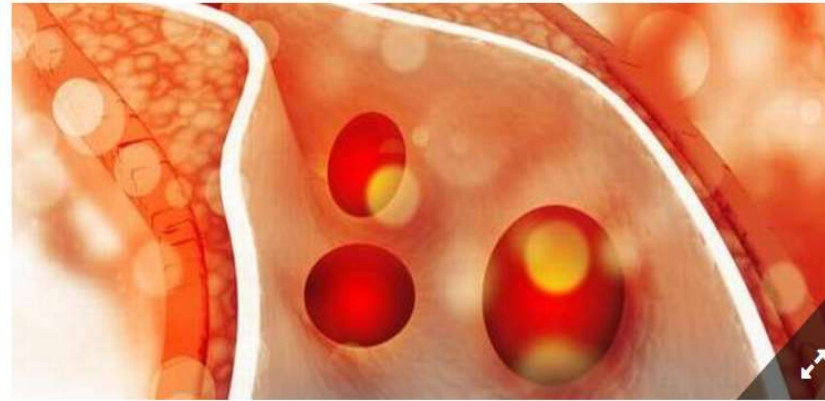
What happens if the forecast indicates that a hurricane is imminent or that tornadoes are likely to occur? Emergency management agencies move into high gear to be ready to rescue people. Emergency teams rely on statistics to tell them when danger may occur

Predicting Disease

Statistics is even playing a role in the medical field. Statistics help us to know how many numbers of people are suffering from the disease. It also helps us to understand how many have died from the same disease.

Cholesterol increases Covid-19 infectivity: Study

Prashasti Awasthi Mumbai | Updated on January 30, 2021 Published on January 30, 2021



It showed that dietary supplements and compounds — vitamin D and possibly vitamins K and A — could bind

to the viral spike protein, reducing SARS-CoV-2 infectivity



Political Campaigns



- Statistics are crucial in a political campaign. Without statistics, no one can run a political campaign with perfection. It helps the politicians to have an idea about how many chances they have to win an election in a particular area.
- Statistics also help the news channel to predict the winner of the election. It also helps the political parties to know how many candidates are in their support in a particular voting zone.

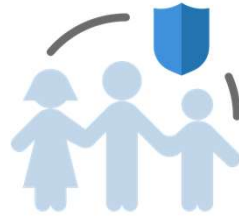
TMC To 'Lose Seats' As Per Party's Survey, Shows Increase In BJP's Share In West Bengal Assembly Elections



According to an internal survey conducted by TMC, in the upcoming elections, BJP may get 98 seats and Mamata Banerjee's vote share might get a spike of little more than 51 per cent.

by: ABP News Bureau | Updated: 4 January 2021, 9:30 PM (IST)

Insurance



Insurance is a vast industry. There are hundreds of insurance i.e. car insurance, bike, life insurance, and many more. The premium of insurance is based on the statistics. Insurance companies use the statistics that are collected from various homeowners, drivers, vehicle registration office, and many more. They receive the data from all these resources and then decide the premium amount.

**Oriental**
insurance

PRITHVI, AGNI, JAL, AKASH
Sab Ki Suraksha Hamare Paas
IRDA REG. NO. 556 CIN - U66010DL1947GOI007158

LOG IN

RENEW ONLINE

HAPPY FAMILY FLOATER PREMIUM RATE CHART

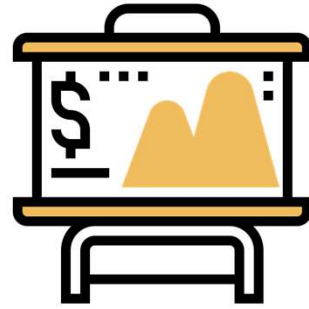
SI/AGE	91 days to 20 yrs	21-35 yrs	36-45 yrs	46-55 yrs	56-60 yrs	61-65 yrs	66-70 yrs	71-75 yrs	76-80 yrs	yrs >80 yrs
SILVER PLAN										
100000	1029	1530	1716	2106	3810	4822	5302	5829	6409	7048
200000	1862	2051	2592	3497	5915	8175	8990	9887	10873	11957
300000	2249	2720	3552	4929	7986	11435	12576	13831	15211	16730
400000	2747	2944	4726	6465	10170	13127	14437	15878	17464	19207
500000	3406	3491	5767	7552	11626	14819	16299	17926	19716	21685

Consumer goods



Wal-Mart, a worldwide leading retailer, keeps track of everything they sell and use statistics to calculate what to ship to each store and when. From analyzing their vast store of information, for example, Wal-Mart decided that people buy strawberry Pop Tarts when a hurricane is predicted in Florida! So they ship this product to Florida stores based upon the weather forecast

Financial Market



The financial market completely relies on the financial market. All the stock prices calculate with the help of statistics. It also helps the investor to take the decision of investment in the particular stock.

Sports



Sports commentators and analysts use statistics to predict the outcome of a game. For instance, the previous five meetings to calculate the possibilities of whether the teams will maintain their form. If a team has won the last five head to head matches against the other, there are high chances of securing that form. Another data that is used to predict the outcome of an incoming game is the health status of a player. Statistics are helpful when determining the chances of a player returning from an injury.

Few definitions:

To understand the following definitions, we will use the following study:

As a quality control expert you want to know the proportion of good computer chips produced by a manufacturing unit.

Populations:

A population is a collection of all distinct individuals or objects or items under study. The number of entities in a population, called the population size.

Example: collection of all computer chips produced by the manufacturing unit.

Variable:

Once a population is fixed, we then want to study a characteristic of the individuals (or objects or items) in the population. This individual characteristic is called a variable (value of which can vary from individual to individual, or from one object to another within the population).

Example: quality of a computer chip.

Most of the variables can be classified in two broad categories:

- (i) **Categorical (or qualitative) variable:** If it assigns a categorical (non-numerical) value to each individual or object. Examples: gender, nationality, etc.
- (ii) **Quantitative variable:** If it assigns a numerical value to each individual or object. Examples: age, income, height, etc.

Example: quality of a computer chip is a categorical variable (since chip is classified as either good or bad).

Parameter:

A summary value of the variable for the population is called a parameter. In a statistical study we are interested in a parameter since often it gives a fairly good idea about the population.

Example: the percentage of good computer chips in the population is a parameter of interest.

Note: A variable is a characteristic of an individual or object, and a parameter is a characteristic of the population (with respect to a variable).

Sample:

A sample is a part of a population and the sample size is denoted by n . A sample should be a representative of the population.

Basic Steps in a Statistical Study

For any statistical study, there are some basic steps to be followed once we draw a sample. These are:

- Step 1: Gather first-hand information from the sample and this is called the raw data.
- Step 2: Tabular representation of the raw data, i.e., represent the raw data in a table.
- Step 3: Pictorial representation of the data, i.e., draw diagrams with the organized data in a table.
- Step 4: Numerically summarize the data, i.e., describe the entire data set with some key numbers.
- Step 5: Analyze the data using mathematical formulae.
- Step 6: Draw the final inference or conclusion about the population under study.

Representation of the data

Given a raw data set, we can rearrange it in two different ways.

❖ **Frequency distribution or Discrete frequency distribution:**

Using the frequency of the variable we can arrange it. This representation of the data is known as frequency distribution.

❖ **Grouped frequency distribution or Continuous frequency distribution:**

Again we can arrange it for the class intervals. For this situation, it is called as Grouped frequency distribution of the variable.

Frequency distribution of ungrouped data:

Given below are marks obtained by 20 students in Math out of 25.

21, 23, 19, 17, 12, 15, 15, 17, 17, 19, 23, 23, 21, 23, 25, 25, 21, 19, 19, 19

Marks Obtained	Tally Marks	Frequency
12	I	1
15	II	2
17	III	3
19	IIII	5
21	III	3
23	IIII	4
25	II	2

Frequency distribution of grouped data:

The presentation of the above data can be expressed into groups. These groups are called classes or the **class interval**.

Each class interval is bounded by two figures called the **class limits**.

Marks	Number of Students (Frequency)
0 - 10	0
10 - 20	11
20 - 30	9

Note: The lower value of a class interval is called lower limit and upper value of that class interval is called the upper limit. Thus, each class interval has lower and upper limits.

For Example:

In the class interval 10 - 20, 10 is the lower limit and 20 is the upper limit.

Exclusive form of data:

The previous table is expressed in the exclusive form.
The class intervals are 0 - 10, 10 - 20, 20 - 30.
In this, we include lower limit but exclude upper limit.
So, 10 - 20 means values from 10 and more but less than 20.
20 - 30 would mean values from 20 and more but less than 30.

Data in the inclusive form:

Marks obtained by 20 students in Math test are given below.
23, 0, 14, 10, 15, 3, 8, 16, 18, 20, 1, 3, 20, 23, 24, 15, 24, 22, 14, 13
Let us represent this data in the inclusive form.

Marks	Number of Students (Frequency)
0 - 10	6
11 - 20	9
21 - 30	5

How to make it continuous?

If “ d ” is the gap between the upper limit of any class and the lower limit of the succeeding class, the class boundaries for any class are then given by:

$$\begin{aligned}\text{Upper class boundary} &= \text{Upper class limit} + \frac{d}{2} \\ \text{Lower class boundary} &= \text{Lower class limit} - \frac{d}{2}\end{aligned}$$

Marks	Number of Students (Frequency)
-0.5 – 10.5	6
10.5 – 20.5	9
20.5 – 30.5	5

Summarizing a raw data set or an organized data set

There are two basic properties of a quantitative data set that are commonly studied. These are central tendency and variability (or dispersion).

Central Tendency:

Quite often it is found that the entries in data set cluster around a central (or middle) value. This behavior of the data set is called the central tendency. The main Challenge is to locate a central value around which the clustering takes place. Three standard methods to measure the location of central tendency are:

- Mean
- Median
- Mode

Variability or Dispersion:

Variability or dispersion of data set means the amount of discrepancies among the data entries. There are several ways to measure dispersion or variability in a data set and these are:

- Range
- Quartile deviation
- Variance
- Standard deviation

Arithmetic Mean or Average

❖ Mean of “ n ” observations $(x_1, x_2, \dots, x_n)$ is given by

$$\bar{x} = \frac{1}{n} \sum x_i .$$

Example: A Monthly income of 10 families in a city is given by:
1600, 1560, 1440, 1530, 1670, 1860, 1750, 1910, 1490, 1800.

Ans:

$$\bar{x} = \frac{1}{10} \sum_{i=1}^{10} x_i = \frac{16610}{10} = 1661.$$

❖ In case of the discrete frequency distribution:

If f_i 's are the frequencies of the variable x_i 's then mean

$$\bar{x} = \frac{1}{N} \sum_{i=1}^n f_i x_i , \quad \text{where } N = \sum_{i=1}^n f_i .$$

Example: The following data represents the marks obtained by 60 students of a class. Obtain the average marks.

Marks	20	30	40	50	60	70
No. of Students	8	12	20	10	6	4

Ans:

Marks	No. of Students (f)	fx
20	8	160
30	12	360
40	20	800
50	10	500
60	6	360
70	4	280
	$N = 60$	$\Sigma fx = 2460$

$$\begin{aligned}\bar{x} &= \frac{1}{N} \sum_{i=1}^n f_i x_i \\ &= \frac{2460}{60} \\ &= 41\end{aligned}$$

❖ In case of the continuous frequency distribution:

If f_i 's are the frequencies of the variable x_i 's then mean

$$\bar{x} = \frac{\sum(\text{midvalue of each class}) \times f_i}{\sum f_i}.$$

Example: Obtain the Arithmetic mean for the following data.

Marks	0-10	10-20	20-30	30-40	40-50	50-60
No. of Students	12	18	27	20	17	6

Ans:

Marks	No. of Students (f)	Mid-point (x)	fx
0-10	12	5	60
10-20	18	15	270
20-30	27	25	675
30-40	20	35	700
40-50	17	45	765
50-60	6	55	330

$$\bar{x} = \frac{2800}{100} = 28$$

Calculation of mean by using deviation concept:

Sometime the values of the variable (x) or frequency (f) or both are large. Then, the calculation of mean by previous formulas is quite time-consuming. Hence, to avoid such situations we are calculating mean by taking the deviations of the given values from any arbitrary point “ A ” as explained below:

Discrete frequency distribution: $\bar{x} = A + \frac{1}{N} \sum_{i=1}^n f_i d_i$,

where “ A ” is an arbitrary point, $d_i = x_i - A$ and $N = \sum_{i=1}^n f_i$.

Continuous frequency distribution: $\bar{x} = A + \frac{h}{N} \sum_{i=1}^n f_i d_i$,

where “ A ” is an arbitrary point, “ h ” is the magnitude of class interval and $N = \sum_{i=1}^n f_i$. Here, $d_i = \frac{x_i - A}{h}$, where x_i ’s are the mid value of each class.

Example:

C.I.	0-8	8-16	16-24	24-32	32-40	40-48
Frequency	8	7	16	24	15	7

Ans:

C.I.	Mid-Value	Frequency (f)	$d = \frac{x - A}{h}$	fd
0-8	4	8	-3	-24
8-16	12	7	-2	-14
16-24	20	16	-1	-16
24-32	28	24	0	0
32-40	36	15	1	15
40-48	44	7	2	14
Total		77		-25

$$\bar{X} = A + \frac{h}{N} \sum_{i=1}^n f_i d_i = 28 + \frac{8 \times (-25)}{77} = 25.404$$

Measure of Central Tendency	Merits	Demerits :
a) Arithmetic Mean	1) It is rigidly defined. 2) It is easy to understand and easy to calculate. 3) It is based on all the observations. 4) It is suitable for further mathematical manipulations. 5) Of all the averages, it is affected least by sampling fluctuations.	1) It is affected very much by extreme values 2) It cannot be found for distributions with open end classes. 3) It cannot be used if measurements are qualitative.

Median

- Median of a distribution is the value of the variable which divides it into two equal parts.
- It is the value which exceeds and is exceeded by the same number of observations. Thus the median is called as a “positional average”.

Evaluation of Median: For ungrouped data,

- (i) Odd number of observations. (middle value after the values has been arranged in ascending or descending order of magnitude)
- (ii) Even number of observations. (arithmetic mean of two middle terms after the values has been arranged in ascending or descending order of magnitude)

Ex: Find the median of the values 25, 20, 15, 35, 18.

Ex: Find the median of the values 8, 20, 50, 25, 15, 30.

For Grouped data:

(i) Discrete Frequency distribution:

1. Find $\frac{N+1}{2}$, where N –Total Frequency $= \sum_{i=1}^n f_i$.
2. See the cumulative frequency (c.f.) just greater than $\frac{N+1}{2}$.
3. The corresponding value of x is median.

Example:

x	1	2	3	4	5	6	7	8	9
f	8	10	11	16	20	25	15	9	6

Calculate the Median of the distribution.

Ans:

x	f	$c.f.$
1	8	8
2	10	18
3	11	29
4	16	45
5	20	65
6	25	90
7	15	105
8	9	114
9	6	120
	$N = 120$	

$$\frac{N + 1}{2} = 60.5$$

The cumulative frequency (c.f.) just greater than $\frac{N+1}{2}$ is 65 and value corresponding to 65 is 5.

$\therefore$ **Median is 5.**

(ii) Continuous Frequency distribution:

In case of continuous frequency distribution, the class corresponding to the c.f. just greater than $N/2$ is called the median class and the value of median is obtained by the following formula:

$$\text{Median} = l + \frac{h}{f} \left(\frac{N}{2} - c \right)$$

Where,

l -is the lower limit of the median class,

f -is the frequency of the median class,

h -is the magnitude of the median class,

c -is the c.f. of the class preceding the median class

Note: The median formula can only be used only for continuous classes without any gaps, i.e., for exclusive type classifications.

Example: Find the median of the following data:

Wages (in Rs.)	2000-3000	3000-4000	4000-5000	5000-6000	6000-7000
No. of workers	3	5	20	10	5

Ans:

<i>Wages (in Rs.)</i>	<i>No. of Employees</i>	<i>c.f.</i>
2000-3000	3	3
3000-4000	5	8
4000-5000	20	28
5000-6000	10	38
6000-7000	5	43
	N = 43	

$$\frac{N}{2} = \frac{43}{2} = 21.5$$

Cumulative frequency just greater than 21.5 is 28 and the corresponding class is 4000-5000. Thus the median class is 4000-5000.

$$l = 4000; h = 1000; f = 20; c = 8$$

$$\text{Median} = 4000 + \frac{1000}{20} (21.5 - 8)$$

$$\therefore \text{Median} = 4675.$$

Example: Find the value of the median from the following data:

No. of days for which absent (less than)	5	10	15	20	25	30	35	40	45
No. of students	29	224	465	582	634	644	650	653	655

Ans: The given cumulative frequency will be converted into ordinary frequency as follows:

<i>Class-Interval</i>	<i>Cumulative frequency</i>	<i>Ordinary frequency</i>
0–5	29	$29 = 29$
5–10	224	$224 - 29 = 195$
10–15	465	$465 - 224 = 241$
15–20	582	$582 - 465 = 117$
20–25	634	$634 - 582 = 52$
25–30	644	$644 - 634 = 10$
30–35	650	$650 - 644 = 6$
35–40	653	$653 - 650 = 3$
40–45	655	$655 - 653 = 2$

Median

$$\begin{aligned} &= 10 + \frac{\frac{655}{2} - 224}{241} \times 5 \\ &= 10 + \frac{103.5 \times 5}{241} = 10 + 2.15 \\ &= 12.15 \end{aligned}$$

State the merits of Median.

- It is easy to calculate and understand.
- It eliminates the effect of extreme values to variable by which it is not affected.
- It is capable of further algebraic treatment for analysing other measures.

What are the limitations of Median

- For calculating median it is necessary to arrange the data, whereas other averages do not need any arrangement.
- Since median is positional average, it's value is not determined by each and every observation.
- The median in some cases cannot be computed exactly, as in the case of mean.i.e When the number of items included in a series of data is even.
- The value of median is affected more by sampling fluctuations.

Mode

Definition: Mode is the value of the variable which is predominant in the series.

1. For a discrete frequency distribution , mode is the value of x –*corresponding to maximum frequency.*

x	1	2	3	4	5	6	7	8
f	4	9	16	25	22	15	7	3

2. For Continuous Frequency distribution:

$$\text{Mode} = l + \frac{h(f_1 - f_0)}{(f_1 - f_0) - (f_2 - f_1)} = l + \frac{h(f_1 - f_0)}{2f_1 - f_0 - f_2}$$

l —lower limit of the modal class, h —magnitude of the modal class

f_1 -frequency of the modal class

f_0 and f_2 -frequencies of the class preceding and succeeding the modal class

Example: Find mode

<i>C.I.</i>	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80
<i>Frequency</i>	5	8	7	12	28	20	10	10

Hence the maximum frequency is 28.

Thus the modal class is 40-50.

$$\text{Mode} = 40 + 10 \frac{(28-12)}{2*28-12-20} = 46.67$$

A distribution is having only one mode is called *Unimodal*.

If it contains more than one mode, it is called *bimodal* or *multimodal*.

Note: In the following three cases, mode can not be obtained by using the above formula:

(a) When the highest frequency is observed at the beginning of the frequency table.

(b) When the highest frequency is observed at the ending of the frequency table.

(c) When two or more class intervals contain the same maximum frequencies.

However, in the above three cases, mode can be obtained by using either a method called ‘Grouping method’ or ‘empirical relationship between arithmetic mean, median and mode.

The empirical relationship between mean, median and mode is

$$\text{Mean} - \text{Mode} = 3 (\text{Mean} - \text{Median})$$

Or,

$$\text{Mode} = 3 \text{ Median} - 2 \text{ Mean.}$$

Calculate the mean, median and mode for the following data.

<i>Wages (in lakhs.)</i>	15-19	20-24	25-29	30-34	35-39	40-44	45-49	50-54	55-59	60-64
<i>No. of workers</i>	31	47	59	78	104	113	81	60	52	25

Mean =39.52

Median=39.77

Mode=40.6

Calculate the Mean, Median and Mode for the following data.

<i>Variable</i>	10-13	13-16	16-19	19-22	22-25	25-28	28-31	31-34	34-37	37-40
<i>Frequency</i>	8	15	27	51	75	54	36	18	9	7

Mean=24.19

Median=23.96

Mode=23.6

Uses of Mode :Mode finds an important place in marketing studies, where a manager of a business concern is interested in knowing about the size which has the highest concentration of items. For example, in placing an order for shoes or ready made garments, the model size helps because this size and other sizes around it are in common demand. It is also used in dealing with non-quantitative data

<i>Series A</i>	<i>Series B</i>	<i>Series C</i>
200	200	1
200	205	989
200	202	2
200	203	3
200	190	5
Mean = 200	200	200

Measure of Dispersion

❖ *Scatteredness (homogeneity or heterogeneity)*

The degree to which numerical data tend to spread about an average value is called variation or dispersion of data.

Measures of Dispersion:

- Range
- Quartile Deviation
- Mean Deviation
- Standard Deviation

Range

Definition: Difference between the value of the smallest item and the value of the largest item in the distribution.

$$\text{Range} = L - S$$

L – Largest Value, S- Smallest Value

The *relative measure* corresponding to range is called the *coefficient of range*,

$$\text{Coefficient of Range} = \frac{L-S}{L+S}$$

Example 1: The following are the prices of shares of a company from Monday to Saturday:

Days	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Price(Rs.)	200	210	208	160	220	250

Calculate the range and its coefficient.

Solution: $\text{Range} = L - S = 250 - 160 = 90$

Range = Rs. 90

$$\text{Coefficient of Range} = \frac{L-S}{L+S} = \frac{250-160}{250+160} = \frac{90}{410} = 0.22$$

In a frequency distribution, range is calculated by taking the difference between the lower limit of the lowest class and the upper limit of the highest class.

Example 2:

<i>Marks</i>	10-20	20-30	30-40	40-50	50-60	60-70
<i>No. of Students</i>	12	18	27	20	17	6

$$\text{Range} = L - S = 70 - 10 = 60$$

$$\text{Coefficient of Range} = \frac{L-S}{L+S} = \frac{70-10}{70+10} = \frac{60}{80} = 0.75$$

Partitions:

These are the values which divided the series into a number of equal parts.

Quartiles: The three points which divided the series in to four equal parts are called quartiles. It is denoted by Q_1, Q_2, Q_3 .

Deciles: The nine points which divided the series in to ten equal parts are called deciles. It is denoted by $D_1, D_2, \dots, D_9$.

Percentiles: The ninety-nine points which divided the series in to hundred equal parts are called percentiles. It is denoted by $P_1, P_2, \dots, P_{99}$.

For discrete frequency distribution:

Quartiles:- $Q_k : \frac{kN}{4}$; Identify the same value in cf list, otherwise find cf just greater than $\frac{kN}{4}$, the corresponding variable is the quartile value. Here, $k = 1, 2, 3$.

Deciles:- $D_k : \frac{kN}{10}$; Identify the same value in cf list, otherwise find cf just greater than $\frac{kN}{10}$, the corresponding variable is the decile value. Here, $k = 1, 2, 3, 4, 5, 6, 7, 8, 9$.

Percentiles:- $P_k : \frac{kN}{100}$; Identify the same value in cf list, otherwise find cf just greater than $\frac{kN}{100}$, the corresponding variable is the percentile value. Here, $k = 1, 2, \dots, 99$.

For continuous frequency distribution:

Quartiles:- Step-1: Find quartile class by:

Compute $\frac{kN}{4}$; Identify the same value in cf list, otherwise
find cf just greater than $\frac{kN}{4}$.

Step-2: Use the formula

$$Q_k = l + \frac{h}{f} \left(\frac{kN}{4} - c \right) ; k = 1, 2, 3.$$

Deciles:- Step-1: Find decile class by:

Compute $\frac{kN}{10}$; Identify the same value in cf list, otherwise
find cf just greater than $\frac{kN}{10}$.

Step-2: Use the formula

$$D_k = l + \frac{h}{f} \left(\frac{kN}{10} - c \right) ; k = 1, 2, \dots, 9.$$

Percentiles:- Step-1: Find percentile class by:

Compute $\frac{kN}{100}$; Identify the same value in cf list, otherwise find cf just greater than $\frac{kN}{100}$.

Step-2: Use the formula

$$P_k = l + \frac{h}{f} \left(\frac{kN}{100} - c \right) ; k = 1, 2, \dots, 99.$$

Quartile Deviation

Definition: Average amount by which the two quartiles differ from the median.

$$\text{Quartile Deviation (Q.D.)} = \frac{Q_3 - Q_1}{2}$$

- The Median $\pm$ Q.D. covers exactly 50 per cent of the observations.
- When Q.D. is very small, it describes high uniformity or small variation of the central 50% items, and a high Q.D. means that the variation among the central items is large.

Relative measure of Q.D.

$$\text{Coefficient of Q.D.} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

It can be used to compare the degree of variation in different distributions.

Example 3: Calculate the value of Q.D. and its coefficient of Q.D. from the following data.

Roll No.	1	2	3	4	5	6	7
Marks	20	28	40	12	30	15	50

Solution: Marks in ascending order 12 15 20 28 30 40 50

$$Q_1 = \text{Size of } \frac{N+1}{4} \text{th item} = \text{Size of } \frac{7+1}{4} = 2^{\text{nd}} \text{ item.}$$

Size of 2nd item is 15. Hence $Q_1 = 15$

$$Q_3 = \text{Size of } 3\left(\frac{N+1}{4}\right) \text{th item} = \text{Size of } 3\left(\frac{7+1}{4}\right) = 6^{\text{th}} \text{ item.}$$

Size of 6th item is 40. Hence $Q_3 = 40$.

$$\therefore Q.D. = \frac{Q_3 - Q_1}{2} = \frac{40 - 15}{2} = 12.5$$

$$\text{Coefficient of } Q.D. = \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{40 - 15}{40 + 15} = 0.455$$

Example 4: Compute the value of Q.D. and its coefficient from the following data.

Marks	10	20	30	40	50	60
No. of Students	4	7	15	8	7	2

Solution:

<i>Marks</i>	<i>Frequency</i>	<i>cumulative frequency</i>
10	4	4
20	7	11
30	15	26
40	8	34
50	7	41
60	2	43

Q_1 = Size of $\frac{N+1}{4}$ *th* item = Size of = 11th item.

Size of 11th item is 20. Hence **$Q_1 = 20$**

Q_3 = Size of $3\left(\frac{N+1}{4}\right)$ *th* item = Size of $3\left(\frac{43+1}{4}\right) = 33^{\text{rd}}$ item.

Size of 33rd item is 40. Hence **$Q_3 = 40$** .

$$\therefore Q.D. = \frac{Q_3 - Q_1}{2} = \frac{40 - 20}{2} = \mathbf{10}$$

$$\text{Coefficient of } Q.D. = \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{40 - 20}{40 + 20} = \mathbf{0.333}$$

Example 4: Compute the value of Q.D. and coefficient of Q.D. from the following data

<i>C.I.</i>	10-20	20-30	30-40	40-50	50-60	60-70	70-80
<i>f</i>	12	19	5	10	9	6	6

Solution:

<i>Marks</i>	<i>Frequency</i>	<i>Cumulative Frequency</i>
10-20	12	12
20-30	19	31
30-40	5	36
40-50	10	46
50-60	9	55
60-70	6	61
70-80	6	67
	<i>N</i> = 67	

$$Q_1 = \text{Size of } \frac{N}{4} \text{ th item} = \text{Size of } \frac{67}{4} = 16.75^{\text{th}} \text{ item.}$$

Q_1 lies in the interval **20-30**

$$Q_1 = l + \frac{\frac{N}{4} - c.f.}{f} \times i \quad l = 20, N/4 = 16.75, c.f. = 12, f = 19, i = 10$$

$$Q_1 = 20 + \frac{\frac{67}{4} - 12}{19} \times 10 = 20 + 2.5 = 22.5$$

Hence $Q_1 = 22.5$

$$Q_3 = \text{Size of } \frac{3N}{4} \text{ th item} = \text{Size of } \frac{3 \times 67}{4} = 50.25^{\text{th}} \text{ item.}$$

Q_3 lies in the class **50-60**.

$$Q_3 = l + \frac{\frac{3N}{4} - c.f.}{f} \times i \quad l = 50, 3N/4 = 50.25, c.f. = 46, f = 9, i = 10$$

$$Q_3 = 50 + \frac{50.25 - 46}{9} \times 10 = 50 + 4.72 = 54.72$$

Hence **$Q_3 = 54.72$**

$$Q.D. = \frac{Q_3 - Q_1}{2} = \frac{54.72 - 22.5}{2} = 16.11$$

$$\text{Coefficient of } Q.D. = \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{54.72 - 22.5}{54.72 + 22.5} = 0.4172$$

Merits: When a frequency distribution contains open end classes, Q.D. is the appropriate measure of dispersion.

Demerits: It ignores remaining 50% of the data.

Mean Deviation

Definition: M.D. is the average difference between the observations in a distribution and the *median* or *mean* of that series.

$$\text{Mean Deviation or M.D.} = \frac{\sum |D|}{N}$$

Where $|D|$ is the deviations from *median* ignoring signs.

For individual observations:

- (i) Compute median of the series.
- (ii) Take deviations of items from median ignoring $\pm$ signs and denote these deviations by $|D|$.
- (iii) Obtain the total of these observations, $|D|$.
- (iv) Divide the total obtained in step (iii) by the number of observations to get the value of *mean deviation*.

Relative Measure of M.D. :

$$\text{Co-efficient of M.D.} = \frac{M.D.}{Median}$$

Example 5: Calculate the mean deviation of the two income groups.

I (Rs.)	II (Rs.)
4000	3000
4200	4000
4400	4200
4600	4400
4800	4600
	4800
	5800

Solution:

Mean deviation : **I group** M.D. = $\frac{\sum |D|}{N}$

Median = $\frac{N+1}{2}$ *th* item = $\frac{5+1}{2} = 3^{\text{rd}}$ item. Size of the 3rd item = 4400.

Group I		Group II	
Rs.	<i>D</i>	Rs.	<i>D</i>
4000	400	3000	1400
4200	200	4000	400
4400	0	4200	200
4600	200	4400	0
4800	400	4600	200
		4800	400
		5800	1400
<i>N</i> = 5	 <i>D</i> = 1200	<i>N</i> = 7	 <i>D</i> = 4000

$$\text{M.D.} = \frac{1200}{5} = \mathbf{240}$$

i.e., the average deviation of the individual incomes from the median income is **Rs. 240**.

Mean deviation : **II group** $M.D. = \frac{|D|}{N}$

$$\text{Median} = \frac{N+1}{2} \text{th item} = \frac{7+1}{2} = 4^{\text{th}} \text{ item.}$$

Size of the 4th item = 4400.

$$M.D. = \frac{4000}{5} = 571.43$$

i.e., the average deviation of the individual incomes from the median income is **Rs. 571.43**.

$$\text{Co-efficient of M.D. (I - Group)} = \frac{M.D.}{Median} = \frac{240}{4400} = 0.055$$

$$\text{(II- Group)} = \frac{571.43}{4400} = 0.13$$

Mean deviation – Discrete Series

$$\text{M.D.} = \frac{\sum f|D|}{N}$$

- (i) Compute median of the series.
- (ii) Take deviations of items from median ignoring $\pm$ signs and denote these deviations by $|D|$.
- (iii) Multiply these deviations by the respective frequencies and Obtain the total, $\sum f|D|$
- (iv) Divide the total obtained in step (iii) by the number of observations to get the value of *mean deviation*.

Example 6: The number of telephone calls received at an exchange in 245 successive one-minute intervals are shown in the following frequency distribution. Compute the mean deviation about the median.

<i>Number of Calls</i>	0	1	2	3	4	5	6	7
<i>Frequency</i>	14	21	25	43	51	40	39	12

Solution:

<i>No. of Calls</i>	<i>f</i>	<i>c.f.</i>	<i> D </i>	<i>f D </i>
0	14	14	4	56
1	21	35	3	63
2	25	60	2	50
3	43	103	1	43
4	51	154	0	0
5	40	194	1	40
6	39	233	2	78
7	12	245	3	36
	<i>N = 245</i>			<i>f D = 366</i>

Median = Size of $\frac{N+1}{2}$ th item = $\frac{245+1}{2} = 123^{\text{rd}}$ item .

Hence the median value is 4.

$$\text{M.D.} = \frac{\sum f|D|}{N} = \frac{366}{245} = \mathbf{1.49}$$

In Continuous Series:

We have to obtain the mid-points of the various classes and take the deviations of these **mid-points** from median.

Calculate the coefficient of mean deviation from the following data:

<i>Marks</i>	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90
<i>No. of Students</i>	2	6	12	18	25	20	10	7

Mean deviation = 12.94

Standard deviation

(i) For the ungrouped data:

$$\text{Variance} = \sigma^2 = \frac{\sum (x_i - \bar{x})^2}{N} = \frac{1}{N} \sum x_i^2 - \left(\frac{1}{N} \sum x_i \right)^2 = \frac{1}{N} \sum x_i^2 - \bar{x}^2$$

(ii) Discrete or Continuous frequency distribution

$$\begin{aligned} \sigma^2 &= \frac{1}{N} \sum_i f_i (x_i - \bar{x})^2 \\ &= \frac{1}{N} \sum f_i x_i^2 - \left(\frac{1}{N} \sum f_i x_i \right)^2 \text{ or } \frac{1}{N} \sum f_i d_i^2 - \left(\frac{1}{N} \sum f_i d_i \right)^2 \end{aligned}$$

(Discrete case) (Continuous case)

$$h^2 \left[\frac{1}{N} \sum f_i d_i^2 - \left(\frac{1}{N} \sum f_i d_i \right)^2 \right]$$

$$\text{Standard deviation} = \sqrt{\text{variance}} \quad \sigma = \sqrt{\frac{1}{N} \sum (x_i - \bar{x})^2} \quad \text{or} \quad \sqrt{\frac{1}{N} \sum f_i (x_i - \bar{x})^2}$$

$\bar{x}$ – Arithmetic mean of the distribution

Recall:

$$\bar{x} = A + \frac{1}{N} \sum_{i=1}^n f_i d_i \text{ (mean for discrete), } \bar{x} = A + \frac{h}{N} \sum_{i=1}^n f_i d_i, \text{ (mean for continuous)}$$

(2) Relations between measures of dispersion

(i) *Quartile deviation = $2/3$ (standard deviation)*

(ii) *Mean deviation = $4/5$ (standard deviation)*

Coefficient of Variation

Coefficient of Variation : $C.V. = \frac{\sigma}{\bar{x}} \times 100$ (Relative Measure)

For comparing the variability of two series, we calculate the co-efficient of variations for each series. The series having greater C.V. is said to be more variable than the other and the series having lesser C.V. is said to be more consistent (or homogenous) than the other.

- 3) The score of two players A and B in ten innings during a certain season are:

A	32	28	47	63	71	39	10	60	96	14
B	19	31	48	53	67	90	10	62	40	80

Find which of the two players A, B is more consistent in scoring.

Solution: Calculation of Coefficient of Variation

X	$(X - \bar{X})$	$(X - \bar{X})^2$
32	-14	196
28	-18	324
47	+1	1
63	+17	289
71	+25	625
39	-7	49
10	-36	1296
60	+14	196
96	+50	2500
14	-32	1024
$\sum X = 460$	0	6500

$$\bar{X} = \frac{460}{10} = 46$$
$$\sigma_A^2 = \frac{\sum (x_i - \bar{x})^2}{N} = \frac{6500}{10} = 650$$

Y	$(Y - \bar{Y})$	$(Y - \bar{Y})^2$
19	-31	961
31	-19	361
48	-2	4
53	+3	9
67	+17	289
90	+40	1600
10	-40	1600
62	+12	144
40	-10	100
80	+30	900
$\sum Y = 500$	0	5968

$$\bar{Y} = \frac{500}{10} = 50$$
$$\sigma_B^2 = \frac{\sum (y_i - \bar{y})^2}{N} = \frac{5968}{10} = 596.8$$

$$\sigma_A = \sqrt{\frac{\sum(x_i - \bar{x})^2}{N}} = 25.5$$

$$\sigma_B = \sqrt{\frac{\sum(y_i - \bar{y})^2}{N}} = 24.43$$

$$\text{C.V.}_{(A)} = \frac{\sigma_A}{\bar{x}} \times 100 = 55.43$$

$$\text{C.V.}_{(B)} = \frac{\sigma_B}{\bar{y}} \times 100 = 48.86$$

Example:

Calculate the mean and standard deviation for the following table giving the age distribution of 542 members.

Age in years : 20—30 30—40 40—50 50—60 60—70 70—80 80—90.

No. of members : 3 61 132 153 140 51 2

Solution: Also find CV.

$$\text{Here we take } d = \frac{x - A}{h} = \frac{x - 55}{10}$$

Age group	Mid-value (x)	Frequency (f)	$d = \frac{x - 55}{10}$	fd	fd ²
20 — 30	25	3	-3	-9	27
30 — 40	35	61	-2	-122	244
40 — 50	45	132	-1	-132	132
50 — 60	55	153	0	0	0
60 — 70	65	140	1	140	140
70 — 80	75	51	2	102	204
80 — 90	85	2	3	6	18
		$N = \Sigma f = 542$		$\Sigma fd = -15$	$\Sigma fd^2 = 765$

$$\bar{x} = A + h \frac{\Sigma fd}{N} = 55 + \frac{10 \times (-15)}{542} = 55 - 0.28 = 54.72 \text{ years.}$$

$$\sigma^2 = h^2 \left[\frac{1}{N} \Sigma fd^2 - \left(\frac{1}{N} \Sigma fd \right)^2 \right] = 100 \left[\frac{765}{542} - (0.28)^2 \right]$$

$$= 100 \times 1.333 = 133.3$$

$$\sigma \text{ (standard deviation)} = 11.55 \text{ years}$$

$$CV = 100 \times \frac{\sigma}{\bar{x}}$$

Moments

The r th moment of a variable x about any point $x = A$, usually denoted by μ_r' is given by

$$\begin{aligned}\mu_r' &= \frac{1}{N} \sum_i f_i (x_i - A)^r, \quad \sum_i f_i = N \\ &= \frac{1}{N} \sum_i f_i d_i^r,\end{aligned}$$

where $d_i = x_i - A$.

The r th moment of a variable about the mean $\bar{x}$, usually denoted by μ_r is given by

$$\mu_r = \frac{1}{N} \sum_i f_i (x_i - \bar{x})^r = \frac{1}{N} \sum_i f_i z_i^r$$

where

$$z_i = x_i - \bar{x}.$$

$$\mu_r' = \frac{1}{N} \sum_{i=1}^n f_i (x_i - A)^r$$

$$A = \bar{x}$$

$$\mu_r = \frac{1}{N} \sum_{i=1}^n f_i (x_i - \bar{x})^r$$

Particular Cases

In particular

$$\mu_0 = \frac{1}{N} \sum_i f_i (x_i - \bar{x})^0 = \frac{1}{N} \sum_i f_i = 1$$

and $\mu_1 = \frac{1}{N} \sum_i f_i (x_i - \bar{x}) = 0$, being the algebraic sum of deviations from the mean. Also

$$\mu_2 = \frac{1}{N} \sum_i f_i (x_i - \bar{x})^2 = \sigma^2$$

These results, viz., $\mu_0 = 1$, $\mu_1 = 0$, and $\mu_2 = \sigma^2$, are of fundamental importance and should be committed to memory.

Pearson's Co-efficients

Karl Pearson defined the following

four coefficients, based upon the first four moments about mean :

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3}, \quad \gamma_1 = + \sqrt{\beta_1} \quad \text{and} \quad \beta_2 = \frac{\mu_4}{\mu_2^2}, \quad \gamma_2 = \beta_2 - 3$$


Skewness

Literally, skewness means '*lack of symmetry*'. We study skewness to have an idea about the shape of the curve which we can draw with the help of the given data. A distribution is said to be skewed if

- (i) Mean, median and mode fall at different points,
i.e., $\text{Mean} \neq \text{Median} \neq \text{Mode}$,
- (ii) Quartiles are not equidistant from median, and
- (iii) The curve drawn with the help of the given data is not symmetrical but stretched more to one side than to the other.

Measures of Skewness

Measures of Skewness. Various measures of skewness are

$$(1) S_k = M - M_d \quad (2) S_k = M' - M_0,$$

where M is the mean, M_d , the median and M_0 , the mode of the distribution.

$$(3) S_k = (Q_3 - M_d) - (M_d - Q_1).$$

These are the absolute measures of skewness. As in dispersion, for comparing two series we do not calculate these absolute measures but we calculate the relative measures called the *co-efficients of skewness* which are pure numbers independent of units of measurement.

Co-efficients of Skewness

1. *Prof. Karl Pearson's Coefficient of Skewness.*

$$S_k = \frac{(M - M_0)}{\sigma}$$

II. *Prof. Bowley's Coefficient of Skewness.* Based on quartiles,

$$S_K = \frac{(Q_3 - M_d) - (M_d - Q_1)}{(Q_3 - M_d) + (M_d - Q_1)} = \frac{Q_3 + Q_1 - 2M_d}{Q_3 - Q_1}$$

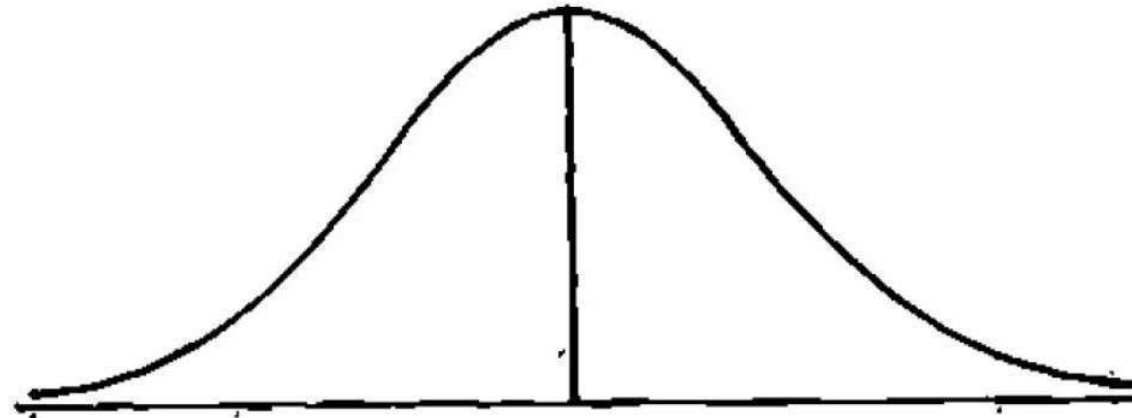
III. Based upon moments, co-efficient of skewness is

$$S_k = \frac{\sqrt{\beta_1} (\beta_2 + 3)}{2 (5\beta_2 - 6\beta_1 - 9)}$$

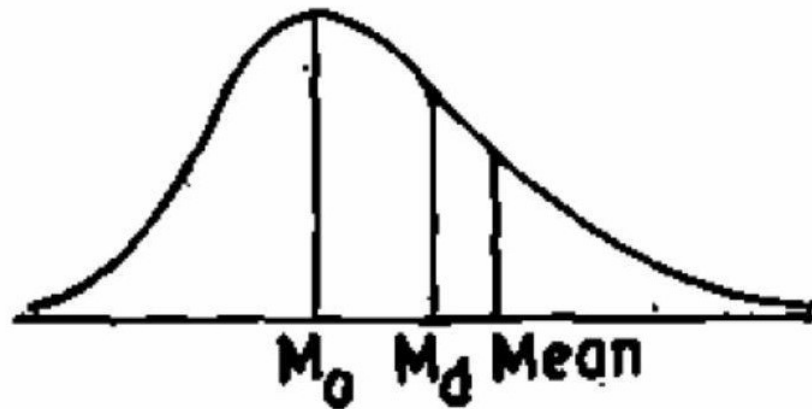
where symbols have their usual meaning. Thus $S_k = 0$ if either $\beta_1 = 0$ or $\beta_2 = -3$. But since $\beta_2 = \mu_4/\mu_2^2$, cannot be negative, $S_k = 0$ if and only if $\beta_1 = 0$. Thus for a symmetrical distribution $\beta_1 = 0$. In this respect β_1 is taken to be a measure of skewness.

The skewness is positive if the larger tail of the distribution lies towards the higher values of the variate (the right), i.e., if the curve drawn with the help of the given data is stretched more to the right than to the left and is negative in the contrary case.

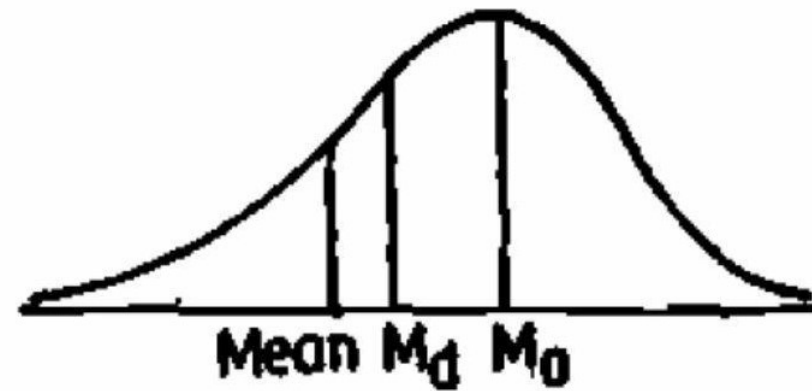
Graphical Representation of Skewness



$\bar{x}$ (Mean) = M_0 = M_d
(Symmetrical Distribution)



(Positively Skewed Distribution)



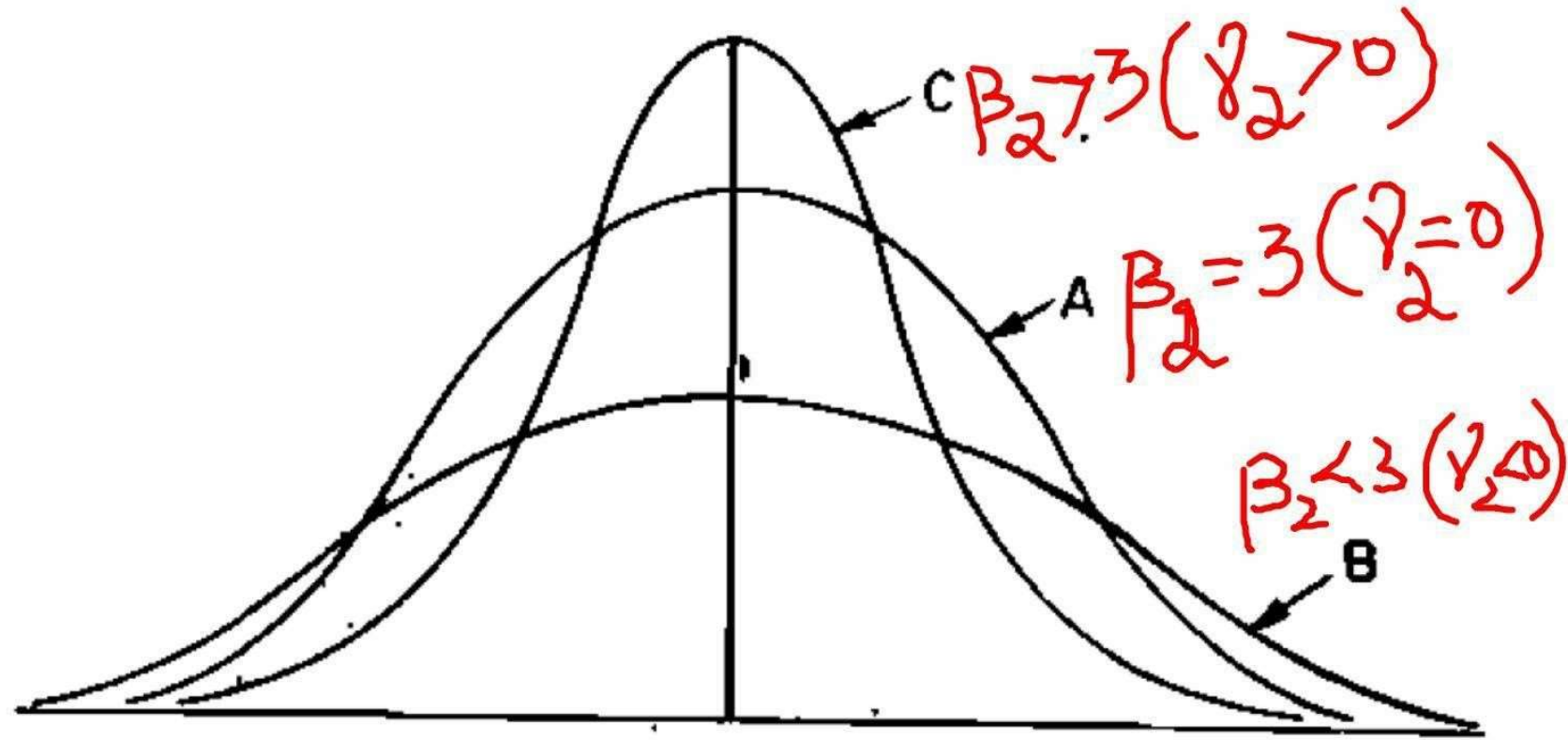
(Negatively Skewed Distribution)

Kurtosis

If we know the measures of central tendency, dispersion and skewness, we still cannot form a complete idea about the distribution as will be clear from the following figure in which all the three curves A, B and C are symmetrical about the mean 'm' and have the same range.

In addition to these measures we should know one more measure which Prof. Karl Pearson calls as the 'Convexity of curve' or *Kurtosis*. Kurtosis enables us to have an idea about the flatness or peakedness of the curve. It is measured by the co-efficient β_2 or its derivation γ_2 given by

$$\underline{\beta_2 = \mu_4/\mu_2^2, \quad \gamma_2 = \beta_2 - 3}$$



Curve of the type 'A' which is neither flat nor peaked is called the *normal curve* or *mesokurtic curve* and for such a curve $\beta_2 = 3$, i.e., $\gamma_2 = 0$. Curve of the type 'B' which is flatter than the normal curve is known as *platykurtic* and for such a curve $\beta_2 < 3$, i.e., $\gamma_2 < 0$. Curve of the type 'C' which is more peaked than the normal curve is called *leptokurtic* and for such a curve $\beta_2 > 3$, i.e., $\gamma_2 > 0$.



Problems based on the Moments, Skewness and Kurtosis concepts:

Q1) Find the first four moments about $x=10$ for the series 4,7,10,13,16,19,22.

Q2) Calculate the first four moments about the mean for the series 4,7,10,13,16,19,22.

Q3) The first four moments of a distribution about $x=4$ are 1, 4, 10 and 45. Comment upon the nature of the distribution.

Q4) In a certain distribution, the first four moments about $x=5$ are 2, 20, 40 and 50. Calculate and state whether the distribution is leptokurtic or platykurtic.

Q5) The first four central moments of a distribution are 0, 2.5, 0.7 and 18.75. Test the skewness and kurtosis of the distribution.